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Paleomagnetism of the Gowganda and Chibougamau Formations: Evidence for 2,200-m.y.-old folding and remagnetization event of the southern province

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ABSTRACT

Paleomagnetic analysis of 175 cores from three regions in the Huronian Gowganda Formation of Ontario and from one area in the probably equivalent Chibougamau Formation of Quebec yielded two remanence directions. Remanence *A* with a pole at lat 66°N, long 035°E (A_{95} = 5°) predates folding of the Huronian rocks; remanence *B* (pole at lat 74°N, long 209°E, A_{95} = 6°) is posttectonic. Together with other Aphebian paleomagnetic results, these data indicate that an event of folding and subsequent remagnetization affected the whole region from Quebec to Minnesota prior to intrusion of the Nipissing diabase at approximately 2,150 m.y. B.P. Similar to other major events in the Canadian Shield, this event is represented in the paleomagnetic record by a closed loop, designated hairpin 50.

INTRODUCTION

The Huronian Supergroup of Ontario and its proposed correlatives in Quebec were extensively deformed, metamorphosed, and subsequently intruded by a suite of (Nipissing) diabase dikes and sills. There are two interpretations of the extent of pre-Nipissing folding of the Huronian rocks: (1) A major orogenic pulse affected the whole southern province prior to Nipissing intrusion (Young and Church, 1966; Church, 1968) or (2) pre-Nipissing deformation was of limited local significance (Brocoum and Dalziel, 1974). Paleomagnetic data from the Gowganda Formation (uppermost Huronian) bearing on these interpretations are presented here.

SAMPLING

Samples were collected from four separate regions: Elliot Lake–Blind River (9 sites—57 cores), Cobalt (7 sites—40 cores), and Whitefish Falls (4 sites—24 cores) in the Gowganda Formation and Lac Waconichi (10 sites—54 cores) in the Chibougamau Formation (Long, 1974). Maps of individual site locations are available (from the author) on request. As many as three specimens were cut from each core. All sites were collected from varvelike argillites. In most cases, the metamorphic grade is lowest greenschist, except for the samples from Whitefish Falls that are of upper-greenschist or lower-amphibolite facies.

Systematic laboratory treatment consisted of stepwise (ten steps) alternating field treatment up to a maximum of 1,000 Oe on one specimen from each core collected. All remaining specimens were thermally cleaned. Occassionally a two-stage cleaning technique was employed: specimens were cleaned first in fields up to 1,000 Oe and then thermally treated up to 675 °C.

RESULTS

Cleaning of the specimens revealed the existence of two remanence directions, *A* and *B* (Fig. 1). Most specimens carried either the *A* or *B* direction. Specimens carrying the *A* direction are from sites (group A) which exhibit little sign of metamorphism. On examination of polished thin sections, the detrital magnetite showed no secondary overgrowths. Magnetically, the specimens carrying only the *A* remanence have little variation of coercivity spectrum (Fig. 2), Koenigsberger ratio (Q_N), or site mean intensity (Fig. 3). The *A* direction, which is found in the Chibougamau, Elliot Lake, and Cobalt regions predates folding of the Huronian Supergroup; fold tests of remanence (Graham, 1949) for the Chibougamau and Elliot Lake regions and for all the results together are positive at the 95% level (Table 1). On the basis of 11 dip-corrected site mean poles, the mean *A* pole is lat 64°N, long 031°E (R = 10.897, k = 99, A_{95} = 5°).

Sites yielding the *B* direction are found from a much broader range of metamorphic grades—from lowest greenschist (Cobalt) to low greenschist (Elliot Lake and Chibougamau) to uppermost greenschist (Whitefish Falls). With the increase of metamorphic grade, the magnetite content increased from an occasional isolated grain with secondary euhedral overgrowths to an anastomosing framework of crystals (Fig. 4). Many of the low-grade metamorphic specimens from the Cobalt region appeared to be stable both to alternating field and thermal cleaning techniques (Fig. 2). However, as shown by Figure 4 this was mainly due to the presence of secondary chemically deposited hematite. Consequently, specimens carrying the *B* direction have a broader range of magnetic properties, coercivity spectrum (Fig. 2), Koenigsberger ratio, and site mean intensity (Fig. 3). Specimens from the higher-grade Whitefish Falls region have an extremely low coercivity spectrum that reflects their large coarse-grain magnetite content, whereas the Cobalt specimens are little affected by alternating frequency cleaning because of the presence of secondary hematite (Figs. 2, 3, 4). Fold tests of remanence for the group *B* directions for the Elliot Lake, Whitefish Falls, and Chibougamau regions and for all sites together are negative; the *B* direction postdates

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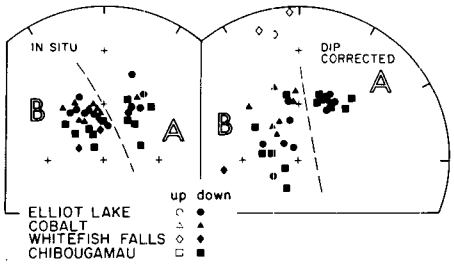


Figure 1. Equal-area projection of site mean directions from Huronian Supergroup and its correlatives. Group A is best grouped after dip correction, whereas group B is grouped better prior to dip correction.

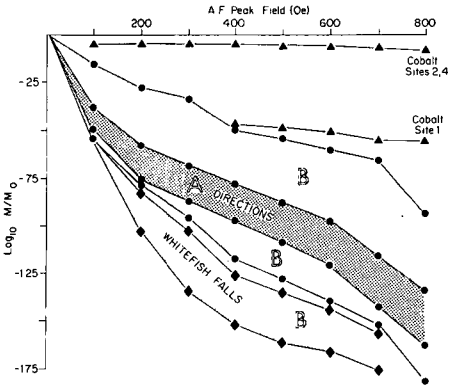


Figure 2. Variation of M/M_0 and its standard deviation with increasing a.f. demagnetizing field. *A* remanence specimens occupy a very restricted field. *B* remanence specimens exhibit a much broader coercivity spectrum. Specimens from Cobalt suggest that hematite could be an important remanence carrier. Whitefish Falls specimens exhibit the lowest coercivity spectrum, which is consistent with the secondary development of metamorphic magnetite.

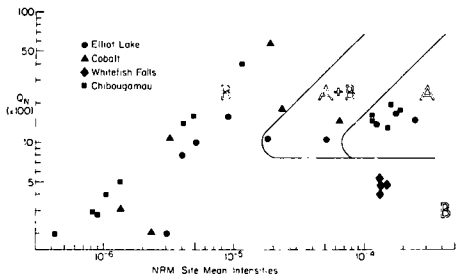


Figure 3. Comparisons of Q_N (Koenigsberger ratio) and site mean intensity. Three fields are recognized that correspond to A, B, and A and B groups. Group A exhibits very limited variation, which reflects limited range of detrital magnetic grains. Group B exhibits wide range, which reflects variation and extent of various remagnetization processes.

folding of the Huronian rocks (*B* directions in Table 1). The small variation of bedding dip in the Cobalt region precluded a satisfactory fold test for this region. On the basis of 23 in situ site mean poles, the mean *B* pole is at lat 74°N, long 209°E ($R = 22.222$, $k = 28$, $A_{95} = 6°$). Confirmation of the secondary (reset) nature of the *B* direction is provided by an identical postfolding magnetization direction (pole GS, Fig. 3) reported by Symons (1975) also from the Gowganda Formation.

Specimens which carried both the *A* and *B* directions have remanence characteristics intermediate between *A* and *B* (Fig. 3). In most instances the *A* and *B* components could be resolved by vector analysis.

Geologic and paleomagnetic relationships indicate that the temporal sequence of directions should be $A \rightarrow B$ for the following reasons: (1) *A* remanence is prefolding; *B* is post-folding. (2) *A* is associated with detrital grains; *B* is associated with secondary grains—either magnetite overgrowths or chemically deposited hematite. (3) Where both *A* and *B* remanences are present, demagnetization characteristics indicate the remanence acquisition sequence was $A \rightarrow B$.

CONCLUSIONS

First, the *A* pole positions from Chibougamau, Cobalt, and Elliot Lake are identical; this indicates that the argillites of the Chibougamau Formation and the argillites of the Gowganda Formation were deposited penecontemporaneously as suggested by Young (1973).

Second, the statistical identity of the *B* poles from Chibougamau, Cobalt, Elliot Lake, *and* Whitefish Falls indicates that the exhumation which reset the magnetization following completion of deformation and metamorphism in each of these regions was also synchronous.

Third, as suggested by Church (1968), the main folding phase of the Huronian must *predate* intrusion of the Nipissing diabase. Igneous contact tests between the Nipissing diabase and Huronian sedimentary rocks are positive (Symons, 1967; Roy and Lapointe, 1976); therefore, both the prefolding and postfolding remanences in the argillites must predate the Nipissing. Furthermore, all fold tests from the Nipissing are negative (Symons, 1970; Symons and Londry, 1975). The diabase was apparently intruded into rocks that were already folded. The whole region from Chibougamau to Sault Ste. Marie therefore was affected by a folding episode prior to magnetization *B* and after magnetization *A*. Comparison of these Gowganda data with other poles for the period 1,900 to 2,300 m.y. B.P. from Laurentia suggests that this folding and exhumation episode occurred between 2,200 and 2,300 m.y. B.P. (Fig. 5; Roy and Lapointe, 1976). This age (2,200 m.y.) is further supported by the recognition of a radiometric resetting event at this time in the Great Lakes region (Van Schmus, 1976).

Any attempt to define a polar path for the period prior to 2,300 m.y. ago that would incorporate the Huronian *A* magnetization must of necessity be speculative (Fig. 6). Reliable, well-dated poles for this interval are scarce. The sequence of these results is based on the radiometric age

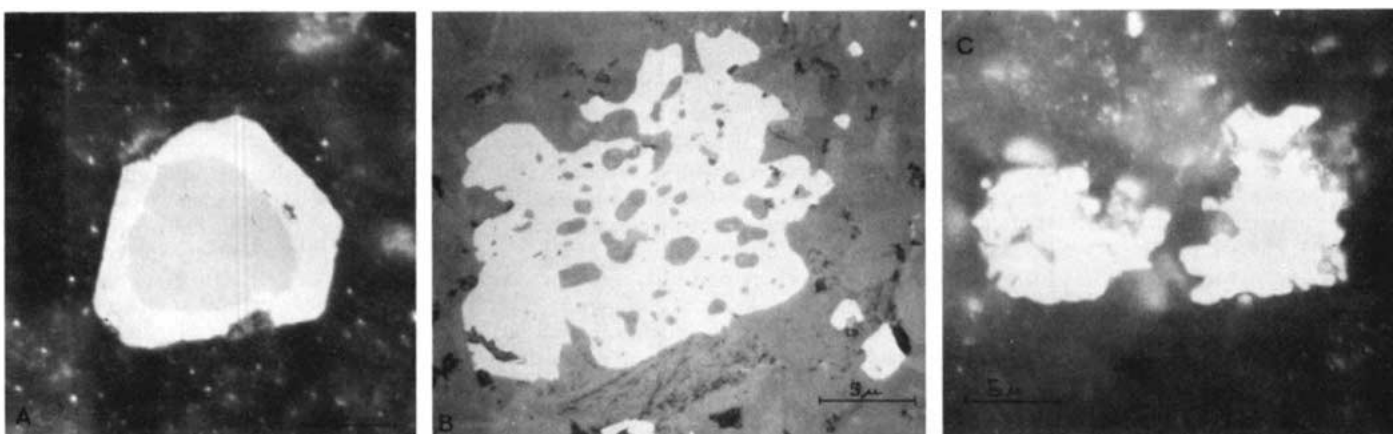


Figure 4. Photomicrographs showing various remagnetization processes. (A) Secondary euhedral overgrowth of magnetite on detrital magnetite, Elliot Lake. (B) Anastomosing grain of secondary magnetite, Whitefish Falls. (C) Secondary hematite grains, Cobalt region.

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TABLE 1. MAGNETIZATION DIRECTIONS

	D	I	N(n)	R	k	α_{95}	Lat (°N)	Long (°E)
			A directions					
Elliot Lake								
Dip corrected	24	54	5(29)	4.986	288.3	15	69	031
In situ	26	54	5(29)	4.936	62.5	10	68	029
Chibougamau								
Dip corrected	35	54	5(26)	4.970	134.7	7	62	035
In situ	39	64	5(26)	4.859	28.5	15	64	009
Cobalt								
Dip corrected	28	54	1(8)	..	..	..	66	030
Overall								
Dip corrected			11*	10.897	98.6	5†	64	031
In situ			11*	10.652	28.7	9†	67	017
			B directions					
Elliot Lake								
In situ	338	68	6(52)	5.943	87.6	7	74	184
Dip corrected	326	74	6(52)	5.796	24.5	14	66	233
Whitefish Falls								
In situ	339	71	4(34)	3.946	56.0	12	74	229
Dip corrected	336	6	4(34)	3.024	3.1	63	..	..
Cobalt								
In situ	338	62	7(52)	6.941	100.9	6	74	182
Dip corrected	336	60	7(52)	6.850	40.0	10	72	180
Chibougamau								
In situ	326	74	6(36)	5.927	68.9	8	69	236
Dip corrected	304	75	6(36)	5.785	23.3	14	58	240
Overall								
In situ			23*	22.222	28.3	6†	74	209
			4§	3.966	87.1	10†	74	210
Dip corrected			23*	19.333	5.1	14†	..	..
			4§	3.012	3.1	..	..	..

Note: D = mean declination; I = mean inclination; N = number of sites, (n) = number of means calculated; R, k, α_{95}, and A_{95} are standard paleomagnetic parameters (Fisher, 1953).

*Unit weight to site poles.

†A_{95}.

§Unit weight to region poles.

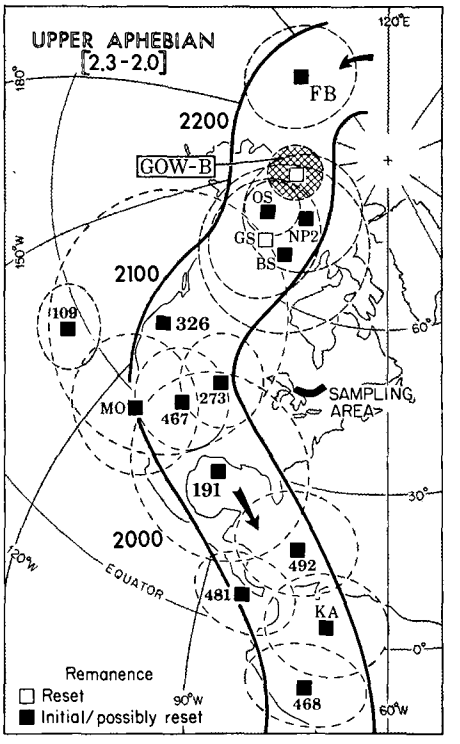


Figure 5. Proposed apparent polar path for Laurentia for the interval 2,300 to 1,900 m.y. B.P. The broad swath (20°) emphasizes the possible uncertainties in determination of paleopoles. Approximate error circles or ovals at P = 0.05 have been drawn. The Gowganda-B pole and the statistically similar GS pole (Symons, 1975) record the same secondary (reset) magnetization. Poles are catalogued by either the three-digit Ottawa number (Irving and Hastie, 1975) or by letter coding: KA, Kahocella Formation; MO, Molson dikes; BS, Big Spruce Complex; OS, Otto stock; NP2, Nipissing; and FB, Firstbrook Formation (all referenced by Roy and Lapointe, 1976).

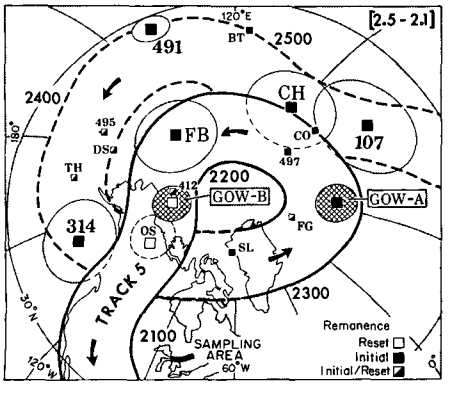


Figure 6. A proposed apparent polar path for Laurentia for the interval 2,500 to 2,100 m.y. B.P. Pole Gowganda-B postdates pole Gowganda-A. The broad swath (20°) emphasizes the possible uncertainties in determination of paleopoles. Approximate error circles or ovals at P = 0.05 have been drawn. The sequence of these results is based on radiometric age assigned to various poles and defines a closed loop–hairpin 50. Poles are catalogued either by the three-digit Ottawa number (Irving and Hastie, 1975) or by letter coding: Primary-quality data–FB, Firstbrook Formation (Roy and Lapointe, 1976); CH, Chibougamau sills (Ueno and Irving, 1976). Secondary data–CO, Coleman member (Roy and Lapointe, 1976); FG, Frechville Greenstone (Halls, 1975); SL, Shelley Lake Granite (Dunlop and Buchan, 1975); TH, Thessalon volcanic rocks (Symons and O'Leary, 1975); DS, Archean volcanic rocks (Dunlop and Buchan, 1975); and BT, Beartooth dike (Larson and others, 1973).

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assigned to the various poles. The *B* magnetization must be predated by the *A* magnetization, which is assigned a Rb-Sr isochron age of 2,288 ± 87 m.y. from apparently unmetamorphosed Gowganda argillite and graywacke (Fairbairn and others, 1969). A Rb-Sr isochron age of 2,369 ± 55 m.y. is accepted as the best age estimate for pole 314 (Fig. 6, Mugford basalts) which therefore must predate pole Gowganda-*A* (Barton, 1975). Poles from the Chibougamau sills (CH, Fig. 6) and Matachewan dikes (107, Fig. 6), dated at approximately 2,500 m.y. B.P., are the final points to define the closed loop. Other poles, which may fall on this loop, are OS (Otto stock) and 491 (Dogrib dikes). Pole OS is plotted on both Figures 5 and 6; two ages of approximately 2,100 (Pullaiah and Irving, 1975) and 2,470 m.y. (Aldrich and Wetherill, 1960) have been assigned to this body. If the latter age is valid, this provides further support for the proposed polar loop. Because there is no contact test on the Dogrib dikes (McGlynn and Irving, 1975), the age of their magnetization is uncertain. It can be accommodated as an initial remanence at approximately 2,600 m.y. B.P. as shown in Figure 6 or as secondary remanence at approximately 1,800 m.y. B.P., as preferred by Roy and Lapointe (1976). In Figure 6, data of poorer quality (whether from lack of age control or limited paleomagnetic reliability) are plotted as small symbols without ovals of confidence. Together, these results suggest that the pole executed a complicated closed loop between 2,100 and 2,500 m.y. B.P.

A diastrophic magnetostratigraphic scheme proposed by Irving and Park (1972) recognized periods of drift (superintervals) and periods of tectonism (hairpins). Subsequent investigations have shown that these hairpins should probably be more correctly termed loops. Similar to other hairpins in the paleomagnetic record of Laurentia, the loop proposed here—referred to as hairpin 50 because it corresponds to the transition from polar track 6 (Irving and Naldrett, 1976, personal commun.) to polar track 5 (Roy and Lapointe, 1976)—coincides with a period of major orogenesis about 2,200 m.y. ago.

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